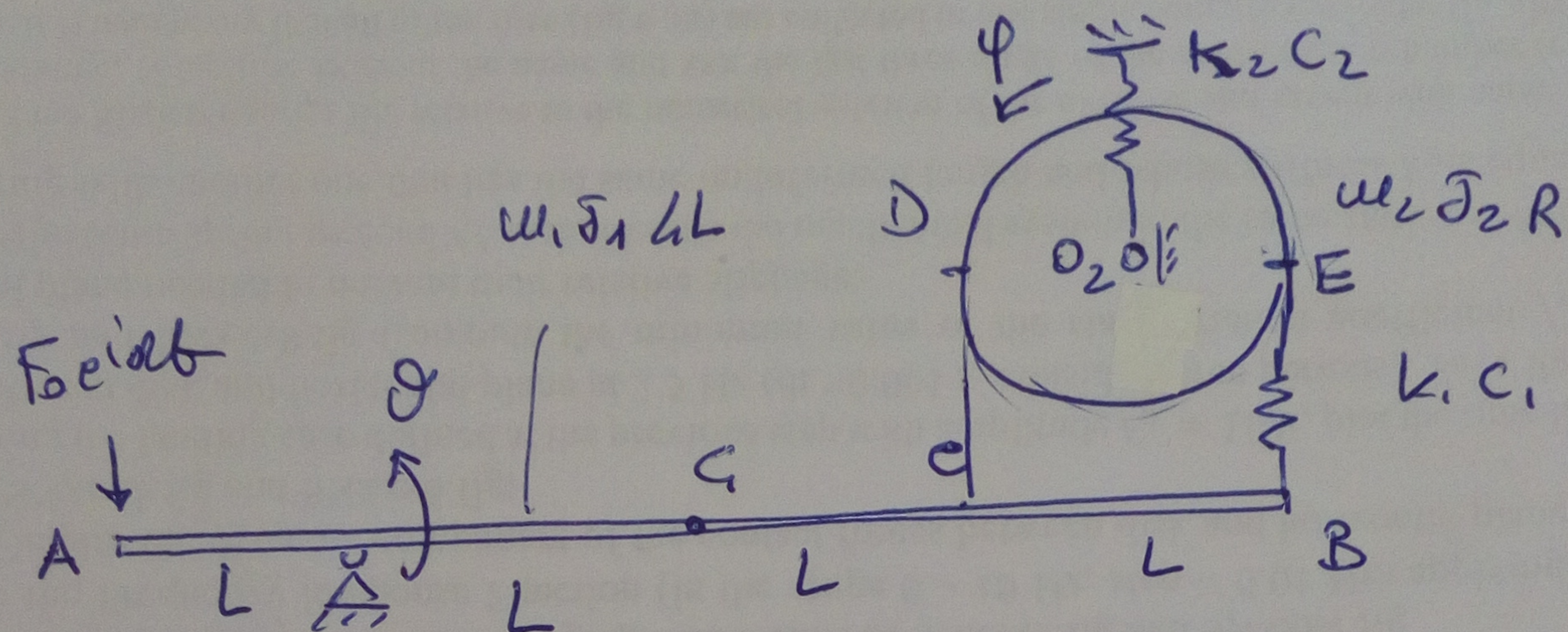


Test June 23rd 2020
-simplified problem-



2 dof $x = \begin{Bmatrix} \theta \\ \phi \end{Bmatrix}$ $x_0 = \begin{Bmatrix} \phi \\ \theta \end{Bmatrix}$

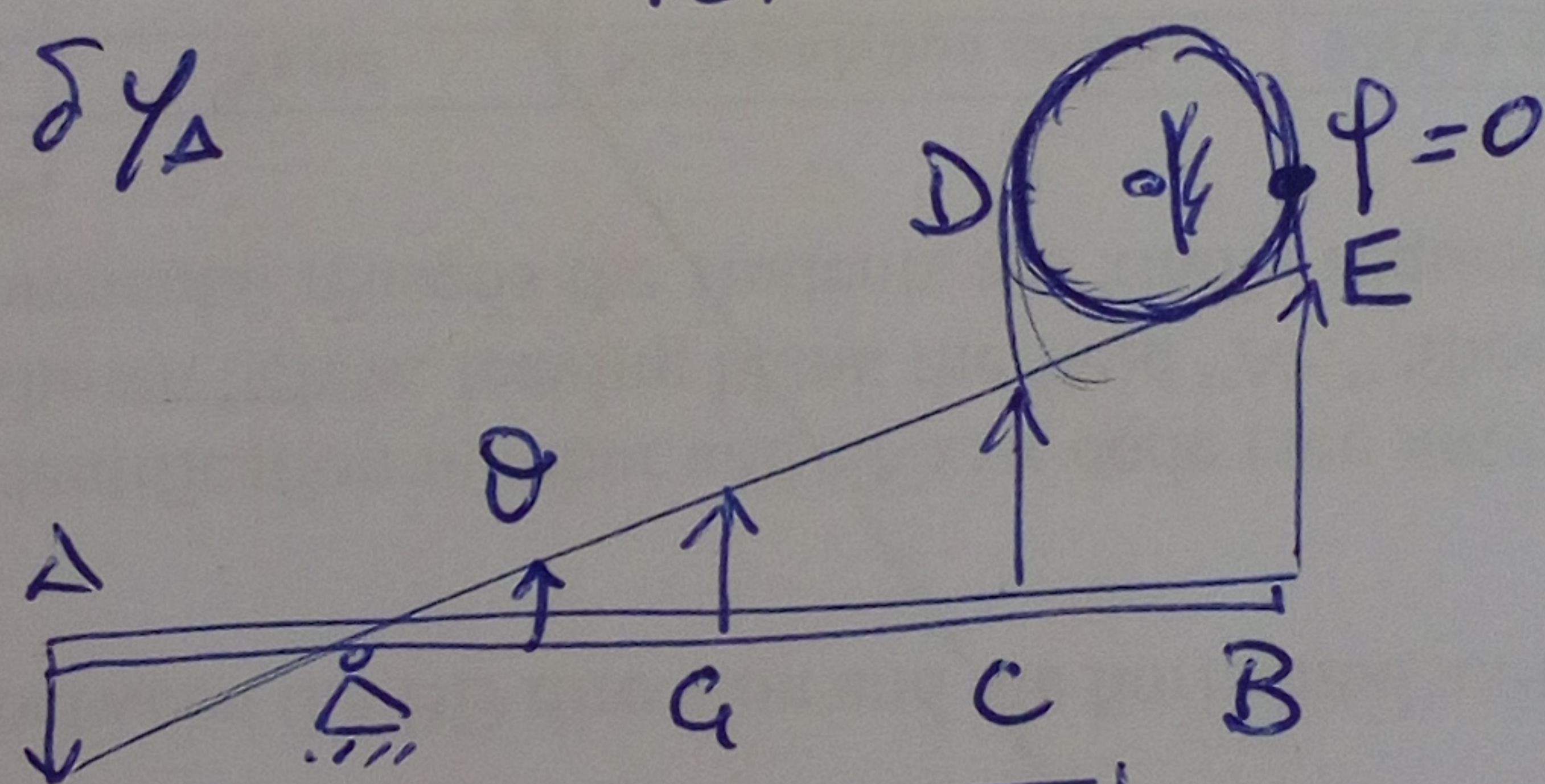
$$T = \frac{1}{2} m_1 \dot{y}_G^2 + \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{y}_{O_2}^2 + \frac{1}{2} I_2 \dot{\phi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^2 C_i \Delta L_i^2 \quad V_{el} = \frac{1}{2} \sum_{i=1}^2 k_i \Delta L_i^2$$

$$V_g = m_1 g h_{G1} - m_2 g h_{G2}$$

$$\delta W = -F_{eiat} \delta y_A$$

$\theta \neq 0; \phi = 0$



$$y_A = -L\theta$$

$$y_G = L\theta$$

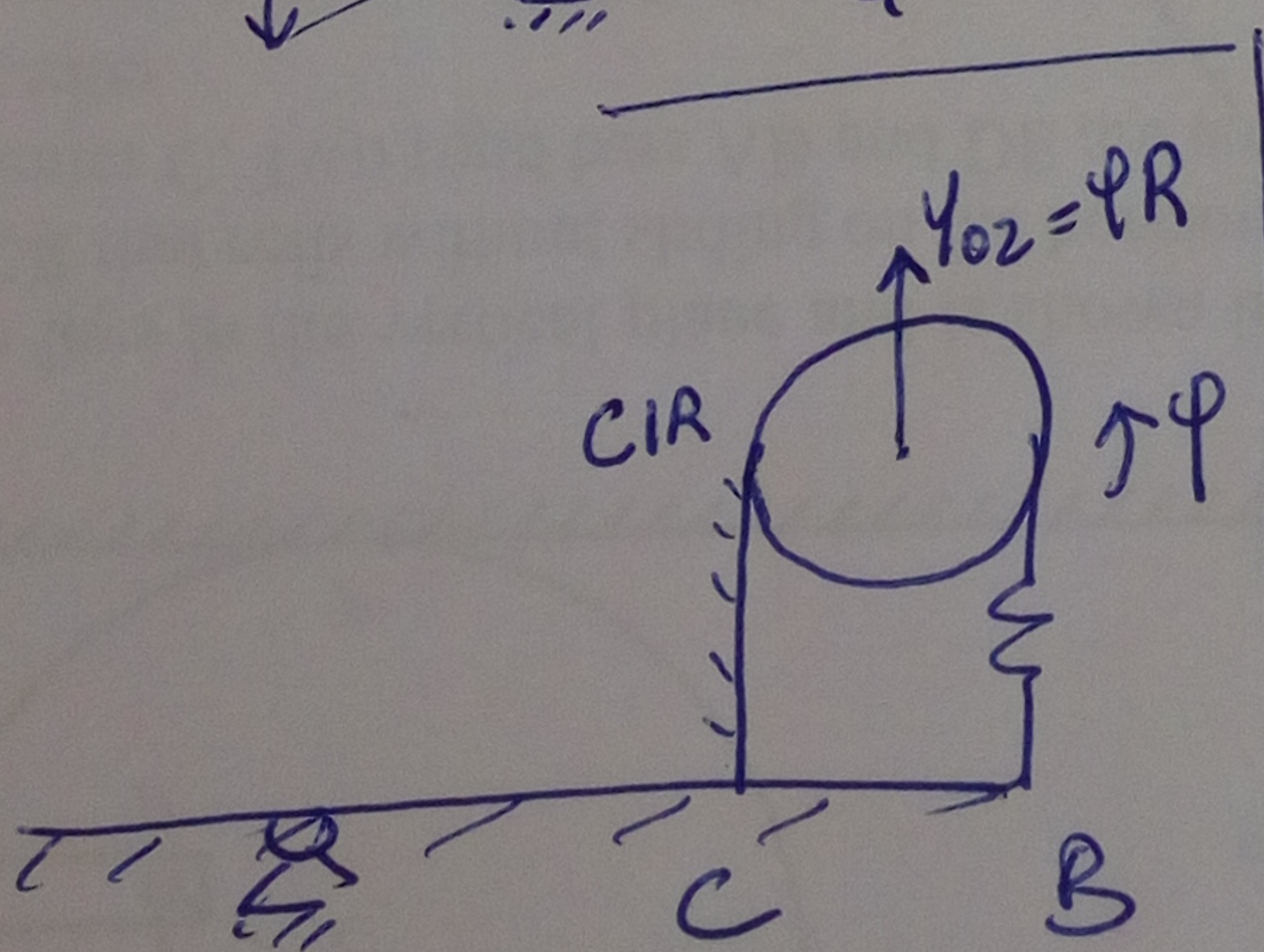
$$y_C = y_D = y_{O_2} = 2L\theta$$

$$y_B = 3L\theta$$

$$\Delta L_1 = y_E - y_B = y_D - y_B = -L\theta$$

$$\Delta L_2 = -y_{O_2} = -2L\theta$$

$\theta = 0; \phi \neq 0$



$$y_A = y_G = y_C = y_B = 0$$

$$y_{O_2} = \phi R \rightarrow \Delta L_2 = -\phi R$$

$$y_E = 2\phi R \rightarrow \Delta L_1 = 2R\phi$$

$$\rightarrow \begin{cases} y_G = L\theta \\ y_{O_2} = 2L\theta + \phi R \\ \Delta L_1 = -L\theta + 2R\phi \\ \Delta L_2 = -2L\theta - \phi R \end{cases}$$

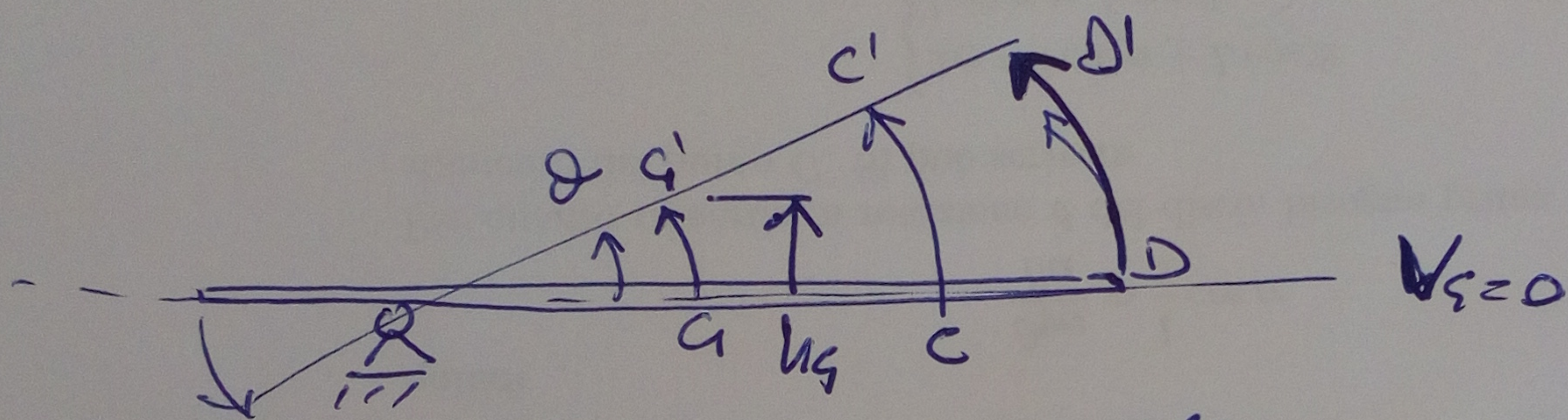
$$y_A = -L\theta$$

$$[u_{PH}] = \text{diag}(u_1, \theta, u_2, \theta) \quad [C_{PH}] = \text{diag}(C_1, C_2)$$

$$[K_{PH}] = \text{diag}(k_1, k_2) \quad \underline{F} = \{F_0\}$$

$$[\Delta u] = \begin{bmatrix} L & 0 \\ 1 & 0 \\ 2L & R \\ 0 & 1 \end{bmatrix} \quad [\Delta K] = [\Delta C] = \begin{bmatrix} -L & 2R \\ -2L & -R \end{bmatrix}$$

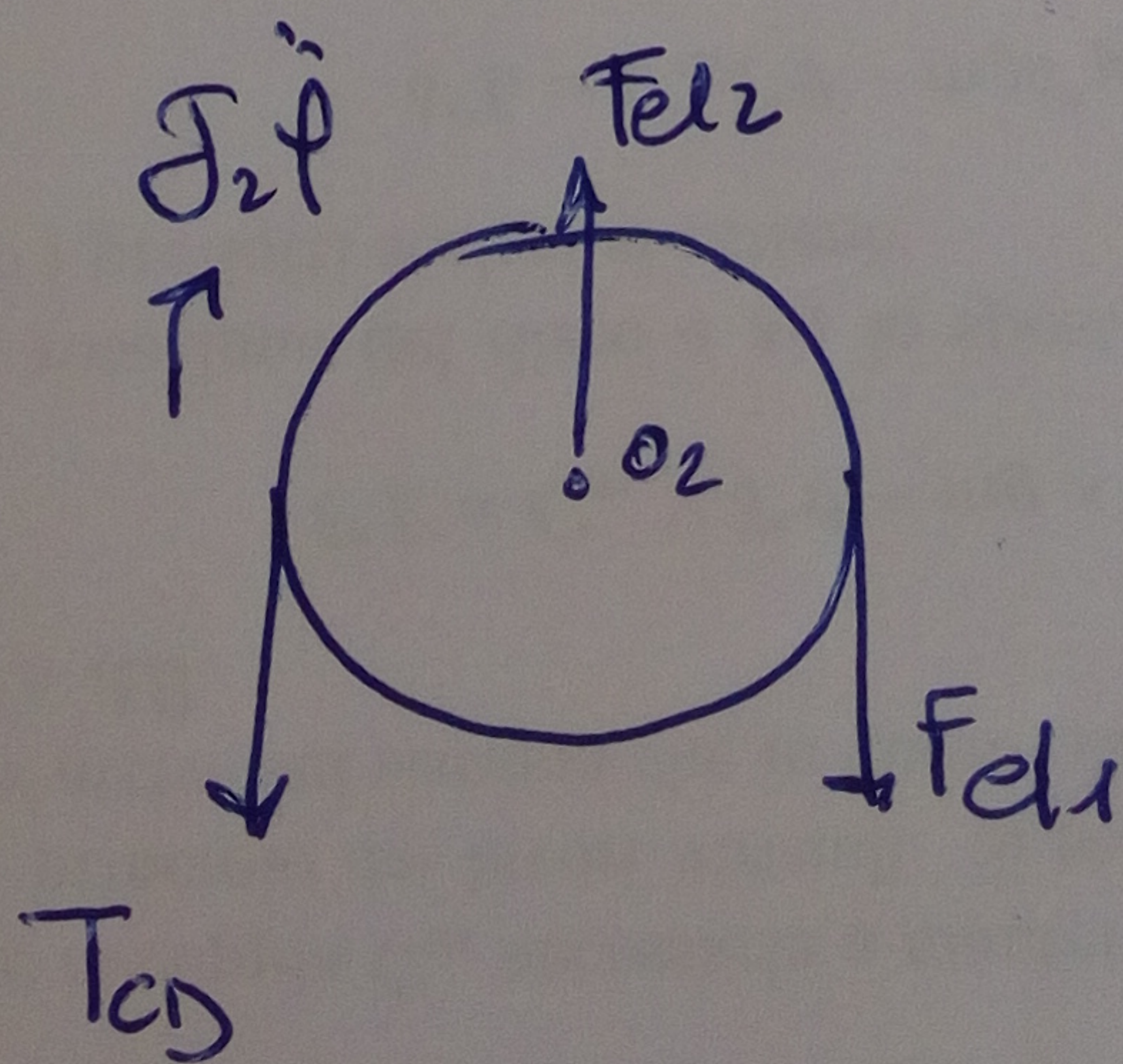
$$[\Delta \phi] = [L \quad \phi]$$



$$h_s = L \sin \theta \rightarrow \frac{\partial^2 h_s}{\partial \theta^2} = -L \sin \theta \rightarrow \frac{\partial^2 h_s}{\partial \theta^2} \bigg|_{\phi} = 0$$

$$[K_s] = \phi$$

- the vertical component of the displacements of points C and B depend on $\sin \theta \rightarrow [K_{EE, \pi}] = 0$



$$\sum M_{O_2}^{\text{disk}} = 0 \quad : \quad F_{e1} R + F_{e2} R - T_{CD} R = 0$$

$$T_{CD} = F_{e1} + F_{e2} R$$